Type MS · Pick the right data type for a value

Answer 1 of 6 · Question 1 part e · 2 marks

2024 JUN · Paper 02 · extract 1 of 16 · find part e in this mark scheme extract

1(a)	A1	**The only correct answer is D** ***A*** *is not correct because it is the definition of decomposition* ***B*** *is not correct because it is a task undertaken during evaluation* ***C*** *is not correct because it describes a method for expressing algorithms*		**1**

Type MS · Pick the right data type for a value

Answer 1 of 6 · Question 1 part e · 2 marks

2024 JUN · Paper 02 · extract 2 of 16 · find part e in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Pick the right data type for a value

Answer 1 of 6 · Question 1 part e · 2 marks

2024 JUN · Paper 02 · extract 3 of 16 · find part e in this mark scheme extract

1b		Award **one** mark for each of:		
	B1	Initialisation of myName to "" (1)	• `String myName = "";` • `String myName = "";` • `myName = ""` Allow null Allow single space between quotes " "	
	B2	Add a line to create myAge and initialise it to 0 (1)	• `int myAge = 0;` • `int myAge = 0;` • `myAge = 0` Allow any method	
	B3	Complete the test by inserting < (1)	• `if (myAge < 30)` • `if (myAge < 30)` • `if (myAge < 30):` Allow <= Can award reverse test only if welcome message contained within an 'else' part	
	B4	Add a line to display "Welcome" (1)	• `Console.WriteLine ("Welcome");` • `System.out.println ("Welcome");` • `print ("Welcome")` ignore indentation Allow any method	**4**

Type MS · Pick the right data type for a value

Answer 1 of 6 · Question 1 part e · 2 marks

2024 JUN · Paper 02 · extract 4 of 16 · find part e in this mark scheme extract

C#

```
// Q01bFINISHED

// Complete the line to initialise myName to an empty string
String myName = "";

// Add a line to create an integer variable called myAge and
//     initialise myAge to 0
int myAge = 0;

Console.WriteLine ("Enter your name: ");
myName = Console.ReadLine();
Console.WriteLine ("Enter your age: ");
myAge = Convert.ToInt32 (Console.ReadLine());

// Complete the test to check if the age is less than 30
if (myAge < 30)
{
    // Add a line to display a message that says "Welcome"
    Console.WriteLine ("Welcome");
}

```

Type MS · Pick the right data type for a value

Answer 1 of 6 · Question 1 part e · 2 marks

2024 JUN · Paper 02 · extract 5 of 16 · find part e in this mark scheme extract

Java

```java
// Q01bFINISHED

import java.util.Scanner;
public class Q01bFINISHED
{
    public static void main (String[] args)
    {
        Scanner myKeyboard = new Scanner (System.in);

        // Complete the line to initialise myName to an empty string
        String myName = "";

        // Add a line to create an integer variable called myAge and
        //     initialise myAge to 0
        int myAge = 0;

        System.out.println ("Enter your name: ");
        myName = myKeyboard.nextLine();
        System.out.println ("Enter your age: ");
        myAge = Integer.parseInt (myKeyboard.nextLine());

        // Complete the test to check if the age is less than 30
        if (myAge < 30)
        {
            // Add a line to display a message that says "Welcome"
            System.out.println ("Welcome");
        }
    }
}
```

Type MS · Pick the right data type for a value

Answer 1 of 6 · Question 1 part e · 2 marks

2024 JUN · Paper 02 · extract 6 of 16 · find part e in this mark scheme extract

Python

```python
# Q01bFINISHED

# Complete the line to initialise myName to an empty string
myName = ""

# Add a line to create an integer variable called myAge and
#     initialise myAge to 0
myAge = 0

myName = input ("Enter your name: ")
myAge = int (input ("Enter your age: "))

# Complete the test to check if the age is less than 30
if (myAge < 30):

    # Add a line to display a message that says "Welcome"
    print ("Welcome")

```

Type MS · Pick the right data type for a value

Answer 1 of 6 · Question 1 part e · 2 marks

2024 JUN · Paper 02 · extract 7 of 16 · find part e in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Pick the right data type for a value

Answer 1 of 6 · Question 1 part e · 2 marks

2024 JUN · Paper 02 · extract 8 of 16 · find part e in this mark scheme extract

1c(i)		Award **one** mark for any of the following:		
	C1	C#: • // Q01c (1) • Q01c (1) • // Loop for 50 times (1) • Loop for 50 times (1) Java: • // Q01c (1) • Q01c (1) • // Loop for 50 times (1) • Loop for 50 times (1) Python: • # Q01c (1) • Q01c (1) • # Loop for 50 times (1) Loop for 50 times (1)	Ignore transcription errors	**1**

Type MS · Pick the right data type for a value

Answer 1 of 6 · Question 1 part e · 2 marks

2024 JUN · Paper 02 · extract 9 of 16 · find part e in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Pick the right data type for a value

Answer 1 of 6 · Question 1 part e · 2 marks

2024 JUN · Paper 02 · extract 10 of 16 · find part e in this mark scheme extract

1c(ii)		Award **one** mark for any of the following:		
	C2	C#: • int count = 0; (1) • int total = 0; (1) • int subtotal = 0; (1) Java: • int count = 0; (1) • int total = 0; (1) • int subtotal = 0; (1) Python: • count = 0 (1) • total = 0 (1) • subtotal = 0 (1)	Ignore transcription errors	1
Question	**mp**	**Answer**	**Additional guidance**	**Mark**

Type MS · Pick the right data type for a value

Answer 1 of 6 · Question 1 part e · 2 marks

2024 JUN · Paper 02 · extract 11 of 16 · find part e in this mark scheme extract

1c(iii)		Award **one** mark for:		
	C3	• if (1)	Applies to all programming languages Ignore capitalisation	1

Type MS · Pick the right data type for a value

Answer 1 of 6 · Question 1 part e · 2 marks

2024 JUN · Paper 02 · extract 12 of 16 · find part e in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Pick the right data type for a value

Answer 1 of 6 · Question 1 part e · 2 marks

2024 JUN · Paper 02 · extract 13 of 16 · find part e in this mark scheme extract

1c(iv)		Award **one** mark for:		
	C4	C#: • && (1) Java: • && (1) Python: • and (1)	Ignore capitalisation	1

Question	mp	Answer	Additional Guidance	Mark

Type MS · Pick the right data type for a value

Answer 1 of 6 · Question 1 part e · 2 marks

2024 JUN · Paper 02 · extract 14 of 16 · find part e in this mark scheme extract

1(d)	D1	**The only correct answer is B** ***A*** *is not correct because the data can be stored in an array as it is all the same data type (string)* ***C*** *is not correct because the data can be stored in an array as it is all the same data type (integer)* ***D*** *is not correct because the data can be stored in an array as it is all the same data type (real)*		**1**

Type MS · Pick the right data type for a value

Answer 1 of 6 · Question 1 part e · 2 marks

2024 JUN · Paper 02 · extract 15 of 16 · find part e in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Pick the right data type for a value

Answer 1 of 6 · Question 1 part e · 2 marks

2024 JUN · Paper 02 · extract 16 of 16 · find part e in this mark scheme extract

<table>
<tr>
<td>1(e)</td>
<td>E1
E2</td>
<td>One mark for each correct cell:
<table>
<tr><th>Item</th><th>Data type</th></tr>
<tr><td>45.82</td><td>All programming languages:
real/float
C#:
decimal/double
Java:
Double/double/BigDecimal
Python:
decimal</td></tr>
<tr><td>True</td><td>All programming languages:
Boolean/bool</td></tr>
</table></td>
<td>Ignore capitalisation
Do not accept string for True</td>
<td>2</td>
</tr>
</table>

Question	mp	Answer	Additional Guidance	Mark

Type MS · Pick the right data type for a value

Answer 2 of 6 · Question 1 part a · 1 mark

2019 JUN · Paper 02 · extract 1 of 10 · find part a in this mark scheme extract

1(a)	**The only correct answer is C** A is not correct because Boolean only has values True and False B is not correct because it only allows for one character D is not correct because it is used for mixed text and numeric data		(1)

Type MS · Pick the right data type for a value

Answer 2 of 6 · Question 1 part a · 1 mark

2019 JUN · Paper 02 · extract 2 of 10 · find part a in this mark scheme extract

1(b)	(an error that) produces incorrect /unintended output (1)	Accept an example of an explained logic error	(1)

Type MS · Pick the right data type for a value

Answer 2 of 6 · Question 1 part a · 1 mark

2019 JUN · Paper 02 · extract 3 of 10 · find part a in this mark scheme extract

1(d)	Any one from: 1. explain how the program works (1) 2. improve readability of the program (1) 3. make code more understandable (1)		(1)

Type MS · Pick the right data type for a value

Answer 2 of 6 · Question 1 part a · 1 mark

2019 JUN · Paper 02 · extract 4 of 10 · find part a in this mark scheme extract

1(e)	1. Global variables created in the main program / can be accessed from any part of the program / are in scope throughout the program (1) 2. Local variables are declared/created inside a function/blocked code (indented, brackets) and are only accessible there / are only in scope within the function in which they're declared (1)		(2)

Type MS · Pick the right data type for a value

Answer 2 of 6 · Question 1 part a · 1 mark

2019 JUN · Paper 02 · extract 5 of 10 · find part a in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Pick the right data type for a value

Answer 2 of 6 · Question 1 part a · 1 mark

2019 JUN · Paper 02 · extract 6 of 10 · find part a in this mark scheme extract

<table>
<tr>
<td>1(g)</td>
<td>For up to two tests
Award 1 mark for an appropriate validation test and 1 mark for an item of test data that would fail only the given test

<table>
<tr><th>Validation test</th><th>Erroneous data</th></tr>
<tr><td>Are the first two characters letters/upper case letters?</td><td>1234CCC</td></tr>
<tr><td>Are the third- and fourth-characters numerical digits?</td><td>CCCCBBB</td></tr>
<tr><td>Are the last three characters letters/upper case letters?</td><td>CC12345</td></tr>
</table>
</td>
<td>Each test should target a different feature of one of the requirements

Exactly 7 characters (in the stem)
First two characters are letters / upper case letters
Characters 3 & 4 are numeric
Last three characters are letters / upper case letters

</td>
<td>(4)</td>
</tr>
</table>

Question	Answer	Additional Guidance	Mark

Type MS · Pick the right data type for a value

Answer 2 of 6 · Question 1 part a · 1 mark

2019 JUN · Paper 02 · extract 7 of 10 · find part a in this mark scheme extract

1(c)	**C#** 1. Semicolon at the end of line 9 [maxValue=5;] (1) 2. Consistent capitalisation of 'count' on lines 10, 11, 13 and 14 (1) 3. constantValue needs to be defined and given the value 7 / replaced by value 7 (1) **Java** 1. Semicolon at the end of line 8 [maxValue = 5;] (1) 2. Consistent capitalisation of 'count' on lines 7, 9, 11 and 12 (1) 3. constantValue needs to be defined and given the value 7 / replaced by 7 (1) **Python** 1. Colon needed at end of line 7 [count, maxValue):] (1) 2. Consistent capitalisation of 'count' on line 3, 7, 8 and 9 (1) 3. constantValue needs to be defined and given the value 7 / replaced by 7 (1)	Candidates are required to open the file Q01c in the code editor. Amended code should be saved as Q01cFINISHED. Line numbers may have been changed by corrections to code (e.g. insertion of constantValue definition)	(3)

Question	Answer	Additional Guidance	Mark

Type MS · Pick the right data type for a value

Answer 2 of 6 · Question 1 part a · 1 mark

2019 JUN · Paper 02 · extract 8 of 10 · find part a in this mark scheme extract

<table>
<tr>
<td>1(f)</td>
<td>Award 1 mark for each correct pair of relational operators and 1 mark for correct Boolean operator for each condition.

<table>
<tr><th>Condition</th><th>Coding</th></tr>
<tr><td>Shop income more than £5000 or assistant sold at least 10 pairs</td><td>shopIncome > 5000; assistantSales >= 10 (1)
correct OR operator (1)</td></tr>
<tr><td>Shop income at least £2000 and assistant sold at least 5 pairs</td><td>shopIncome >= 2000; assistantSales >= 5 (1)
correct AND operator (1)</td></tr>
</table>

Accept >9, >1999 and >4 for respective >= comparisons</td>
<td>Candidates are required to open the file Q01f in the code editor.
Amended code should be saved as Q01fFINISHED.
Do not penalise candidates who attempt more than the stated requirements.
Do not award Boolean operator mark for single | or & as these are not correct.</td>
<td>(4)</td>
</tr>
</table>

Type MS · Pick the right data type for a value

Answer 2 of 6 · Question 1 part a · 1 mark

2019 JUN · Paper 02 · extract 9 of 10 · find part a in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Pick the right data type for a value

Answer 2 of 6 · Question 1 part a · 1 mark

2019 JUN · Paper 02 · extract 10 of 10 · find part a in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Pick the right data type for a value

Answer 3 of 6 · Question 2 part b.iii · 2 marks

2019 JUN · Paper 02 · extract 1 of 13 · find part b.iii in this mark scheme extract

2(b)(i)	24.0 (1)	Accept 24	(1)

Type MS · Pick the right data type for a value

Answer 3 of 6 · Question 2 part b.iii · 2 marks

2019 JUN · Paper 02 · extract 2 of 13 · find part b.iii in this mark scheme extract

2(b)(ii)	One of: By the rules of precedence multiplication is done before addition. (1) Without the brackets the addition would be done to the total of the multiplication. (1) By the rules of the BODMAS/BIDMAS additions happen after multiplication. (1) newSpeed and intialSpeed have to be added together before multiplying (by time). (1)		(1)

Type MS · Pick the right data type for a value

Answer 3 of 6 · Question 2 part b.iii · 2 marks

2019 JUN · Paper 02 · extract 3 of 13 · find part b.iii in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Pick the right data type for a value

Answer 3 of 6 · Question 2 part b.iii · 2 marks

2019 JUN · Paper 02 · extract 4 of 13 · find part b.iii in this mark scheme extract

2(b)(iii)	Real (1) Because real x integer = real (1)	Accept double / float Accept 'because it could be a number with decimal places'	(2)

Type MS · Pick the right data type for a value

Answer 3 of 6 · Question 2 part b.iii · 2 marks

2019 JUN · Paper 02 · extract 5 of 13 · find part b.iii in this mark scheme extract

2(b)(iv)	27.375 (1)		(1)

Type MS · Pick the right data type for a value

Answer 3 of 6 · Question 2 part b.iii · 2 marks

2019 JUN · Paper 02 · extract 6 of 13 · find part b.iii in this mark scheme extract

Q2(c)(i)	C#: 16 or 20 (1) Java: 16 or 21 (1) Python: 8 or 10 (1)		(1)

Type MS · Pick the right data type for a value

Answer 3 of 6 · Question 2 part b.iii · 2 marks

2019 JUN · Paper 02 · extract 7 of 13 · find part b.iii in this mark scheme extract

Q2(c)(ii)	C#: 29 or 31 (1) Java: 24 or 26 (1) Python: 14 or 15 (1)		(1)

Type MS · Pick the right data type for a value

Answer 3 of 6 · Question 2 part b.iii · 2 marks

2019 JUN · Paper 02 · extract 8 of 13 · find part b.iii in this mark scheme extract

Q2(c)(iii)	Any of: 1. Message (1) 2. startPos (Python) / pos (Java or C#) (1) 3. characters (1)		(1)

Type MS · Pick the right data type for a value

Answer 3 of 6 · Question 2 part b.iii · 2 marks

2019 JUN · Paper 02 · extract 9 of 13 · find part b.iii in this mark scheme extract

Q2(c)(iv)	newMessage (1)	Do not accept outMessage	(1)

Type MS · Pick the right data type for a value

Answer 3 of 6 · Question 2 part b.iii · 2 marks

2019 JUN · Paper 02 · extract 10 of 13 · find part b.iii in this mark scheme extract

Q2(c)(v)	Response starts with 'c' (1) Response has six letters in sequence from 'educational' (1)	Marks can be awarded from Q2(c)(vi)	(2)

Type MS · Pick the right data type for a value

Answer 3 of 6 · Question 2 part b.iii · 2 marks

2019 JUN · Paper 02 · extract 11 of 13 · find part b.iii in this mark scheme extract

Q2(c)(vi)	1. Returns substring / crops a string (1) 2. Located between the two input values / with number of characters identified in the last parameter (characters) and starting at the second parameter (startPos) (1)	Award two marks for: It produces a substring of the message starting at position startPos with length characters Award up to 2 marks for a correct interpretation of the code (e.g 'obtain a newMessage' – is worth 1) Accept a correct generic explanation of what a subprogram does for 1 mark	(2)

Type MS · Pick the right data type for a value

Answer 3 of 6 · Question 2 part b.iii · 2 marks

2019 JUN · Paper 02 · extract 12 of 13 · find part b.iii in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Pick the right data type for a value

Answer 3 of 6 · Question 2 part b.iii · 2 marks

2019 JUN · Paper 02 · extract 13 of 13 · find part b.iii in this mark scheme extract

2(a)	Award 1 mark for each of: 1. set counter to 1 and guess to 0 (1) 2. set answer to random value between 1 and 10 (1) 3. request input of and accept an integer value for guess (1) 4. create a while loop with a correct condition (1) 5. increment the counter (1) 6. use if else selection to determine and display appropriate output message for incorrect guesses (1) 7. request input of and accept another integer value for a guess inside the loop (1) 8. display correct guess and count of guesses (1) 9. compiling without syntax errors (1) 10. coding meets all requirements of question (1)	Candidates are required to open the file Q02a in the code editor. Amended code should be saved as Q02aFINISHED. Logic of algorithm must be followed as set out. Alternatives must address each point. Do not penalise candidates who attempt more than the stated requirements.	(10)

Question	Answer	Additional Guidance	Mark

Type MS · Pick the right data type for a value

Answer 4 of 6 · Question 1 part b · 1 mark

2020 NOV · Paper 02 · extract 1 of 10 · find part b in this mark scheme extract

1 (a)	A1	**C** (equals) (1)		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Pick the right data type for a value

Answer 4 of 6 · Question 1 part b · 1 mark

2020 NOV · Paper 02 · extract 2 of 10 · find part b in this mark scheme extract

1 (b)	B1	Award one mark for any of: • TRUE **and** FALSE (1) • Yes **and** No (1) • T **and** F (1) • 0 **and** 1 (1)	Ignore capitalisation	(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Pick the right data type for a value

Answer 4 of 6 · Question 1 part b · 1 mark

2020 NOV · Paper 02 · extract 3 of 10 · find part b in this mark scheme extract

1 (d)	D1	Award one mark for any of: • (Symbolic name associated with) a value that may be changed (1) • A container used to store data (1) • A data store whose contents can change whilst a program is executing (1)		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Pick the right data type for a value

Answer 4 of 6 · Question 1 part b · 1 mark

2020 NOV · Paper 02 · extract 4 of 10 · find part b in this mark scheme extract

1 (e) (i)	E1	Python 6 C# 12 Java 8		(1)

Type MS · Pick the right data type for a value

Answer 4 of 6 · Question 1 part b · 1 mark

2020 NOV · Paper 02 · extract 5 of 10 · find part b in this mark scheme extract

1 (e) (ii)	E2	Python 3 C# 7 Java 8		(1)

Type MS · Pick the right data type for a value

Answer 4 of 6 · Question 1 part b · 1 mark

2020 NOV · Paper 02 · extract 6 of 10 · find part b in this mark scheme extract

1 (e) (iii)	E3	Award one mark for any of: • amount (1) • x (1) • total (1) • count (1)	Accept listNumbers or numArray	(1)

Type MS · Pick the right data type for a value

Answer 4 of 6 · Question 1 part b · 1 mark

2020 NOV · Paper 02 · extract 7 of 10 · find part b in this mark scheme extract

1 (e) (iv)	E4	message		(1)

4 of 16

Type MS · Pick the right data type for a value

Answer 4 of 6 · Question 1 part b · 1 mark

2020 NOV · Paper 02 · extract 8 of 10 · find part b in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Pick the right data type for a value

Answer 4 of 6 · Question 1 part b · 1 mark

2020 NOV · Paper 02 · extract 9 of 10 · find part b in this mark scheme extract

1 (c)	C1	2.1.5	Delete space in variable name (1)		(3)
	C2	2.1.5	Capitalise O in variable name (1)		
	C3	2.1.5	Correction to print statement (1)		
Code example					
Python					

```
numOne = 25
numTwo = 36
numThree = numOne * numTwo
print (numThree)
```

9 of 16

Type MS · Pick the right data type for a value

Answer 4 of 6 · Question 1 part b · 1 mark

2020 NOV · Paper 02 · extract 10 of 10 · find part b in this mark scheme extract

Question	mp	ref	Answer	Additional Guidance	Mark

Type MS · Pick the right data type for a value

Answer 5 of 6 · Question 1 part c · 2 marks

2022 JUN · Paper 02 · extract 1 of 9 · find part c in this mark scheme extract

1 (a)	A1	**The only correct answer is D** A is not correct because the value can be used more than once B is not correct because the value does not always have to be input C is not correct because the value does not always have to be used in a calculation		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Pick the right data type for a value

Answer 5 of 6 · Question 1 part c · 2 marks

2022 JUN · Paper 02 · extract 2 of 9 · find part c in this mark scheme extract

1 (b)	B1	**The only correct answer is B** A is not correct because this would not aid readability of the code C is not correct because this would not aid readability of the code D is not correct because this would not aid readability of the code		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Pick the right data type for a value

Answer 5 of 6 · Question 1 part c · 2 marks

2022 JUN · Paper 02 · extract 3 of 9 · find part c in this mark scheme extract

1 (c)	C1 C2	**Award 1 mark for each of:** **Data type** / **Example value** char / An example of any single letter, number or symbol real / An example of a number with a decimal point	Accept quotes around the character Real number does not need a number after the decimal point to be awarded the mark	(2)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Pick the right data type for a value

Answer 5 of 6 · Question 1 part c · 2 marks

2022 JUN · Paper 02 · extract 4 of 9 · find part c in this mark scheme extract

1 (d)	D1	**Award up to 2 marks for a description such as:** • Boundary testing uses the most extreme valid data / minimum and maximum valid values (1) whereas erroneous testing uses invalid data (1)		(2)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Pick the right data type for a value

Answer 5 of 6 · Question 1 part c · 2 marks

2022 JUN · Paper 02 · extract 5 of 9 · find part c in this mark scheme extract

1 (e)	E1	Logic/logical		(1)

Type MS · Pick the right data type for a value

Answer 5 of 6 · Question 1 part c · 2 marks

2022 JUN · Paper 02 · extract 6 of 9 · find part c in this mark scheme extract

Question	mp		Additional Guidance	Mark

Type MS · Pick the right data type for a value

Answer 5 of 6 · Question 1 part c · 2 marks

2022 JUN · Paper 02 · extract 7 of 9 · find part c in this mark scheme extract

1 (f)	**Award 1 mark for each of:**			
	F1	Bracket added and no other amendments made to the line (1)		
	F2	== becomes = (1)		
	F3	" moved to after **is** and before , OR , area changed to +str(area) (1) Must still include the print statement		(3)

Code examples

Python

```
3   length = float(input("Enter the first number "))
4
5   area = length * length
6
7   print("The area of the square is" , area)
```

Question	mp	Answer	Additional Guidance	Mark

Type MS · Pick the right data type for a value

Answer 5 of 6 · Question 1 part c · 2 marks

2022 JUN · Paper 02 · extract 8 of 9 · find part c in this mark scheme extract

1 (g)	**Award 1 mark for each of:**			
	G1	At least 1 condition correct (1)		
	G2	At least 1 > or < condition and message match (1)		
	G3	At least 1 output includes letter and stored letter with an appropriate message (1)		
	G4	All conditions and outputs correct (1)		(4)
Code examples				
Python				

```
# Amend the code by completing the if statement
if letter > storedLetter:
    print(letter + " is later in the alphabet than " + storedLetter)
elif letter == storedLetter:
    print(letter + " is the same as " + storedLetter)
else:
    print(letter + " is earlier in the alphabet than " + storedLetter)
```

Type MS · Pick the right data type for a value

Answer 5 of 6 · Question 1 part c · 2 marks

2022 JUN · Paper 02 · extract 9 of 9 · find part c in this mark scheme extract

Question	mp		Additional Guidance	Mark

Type MS · Pick the right data type for a value

Answer 6 of 6 · Question 2 part b.i · 2 marks

2025 JUN · Paper 02 · extract 1 of 5 · find part b.i in this mark scheme extract

2(a)	A1	**The only correct answer is D** ***A*** *is not correct because abstraction is a technique used in computational thinking* ***B*** *is not correct because encryption is a security measure* ***C*** *is not correct because iteration is a programming construct*		**1**

Question	mp	Answer	Additional guidance	Mark
2(b)(i)		Award **one** mark for each correct cell:		
	B1 B2	real/double/float: (1) • pay rate/18.75 (1) • total pay/356.25 (1) • tax/0.2/71.25 (1) • income/285.00 (1) int/integer: (1) • hours/19 (1)	Example must match data type for B2 Award suitable example even if not taken from payslip Accept string and matching example e.g. Hours	**2**

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Type MS · Pick the right data type for a value

Answer 6 of 6 · Question 2 part b.i · 2 marks

2025 JUN · Paper 02 · extract 2 of 5 · find part b.i in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
2(b)(ii)		Award **one** mark for each correct cell:		
	B3	**Input:** • Pay Rate/18.75 (1) • Hours/19 (1)		
	B4	**Process**: • Pay Rate * Hours/18.75 * 19 (1) • Total Pay * Tax/356.25 * 0.2 (1) • Total Pay – Tax/356.25 – 71.25 (1)	Don't need output e.g. total pay, just the calculation part e.g. pay rate * hours: Total pay = pay rate * hours Accept 0.2 or 20% as part of calculation for tax Ok to mix types e.g. Total Pay * 20% Accept descriptions such as *multiply the pay rate by hours (hours etc could be taken from input box/ B3 mark)*	
	B5	**Output:** • Total Pay/356.25 (1) • Tax/71.25 (1) • Income/285 (1)		**3**

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Type MS · Pick the right data type for a value

Answer 6 of 6 · Question 2 part b.i · 2 marks

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Question	mp	Answer	Additional Guidance	Mark
2(c)(i)	C1 C2	Award **one** mark for **one** correct cell: Award **two** marks for all **three** correct cells: • **Normal:** any value between 0-255 e.g. 125 • **Boundary:** 0/255 • **Erroneous:** any negative value/any value 256 or above	Accept -1/256 for boundary data Don't accept non-int types	**2**

Question	mp	Answer	Additional Guidance	Mark
2(c)(ii)		Award **one** mark for any of the following up to a maximum of **four** marks:		
	C3	Input changed to int before comparison (1)		**4**
	C4	Relational operator used correctly to check for valid input (both **>=**0 and **<**256 or equivalent) (1)	Allow: >0 and/or <255	
	C5	'and' logical operator used to combine relational tests (1)	or operator acceptable if candidate rearranges the selection statement and test works for correct code option – check if C4 is now awardable	
	C6	Calling of `getHex(userInput)` (1)	Call must be inside print ( ) Subprogram and argument must be functionally correct Ignore additional variables/strings in print statement	

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Type MS · Pick the right data type for a value

Answer 6 of 6 · Question 2 part b.i · 2 marks

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```
37 # complete the line to convert the user input to integer
38 userInput = int(input("please enter a number between 0 and 255 \n"))
39
40 # complete the test to check the input is between 0 and 255
41 if (userInput >=0) and (userInput <256):
42     # complete the line to call getHex subprogram
43     print(getHex(userInput))
44 else:
45     print("number out of range")
```

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Type MS · Pick the right data type for a value

Answer 6 of 6 · Question 2 part b.i · 2 marks

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Question	mp	Answer	Additional Guidance	Mark